Circuit Safari

A Proposal for UROP/FYP/MComp Projects

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1 Motivation

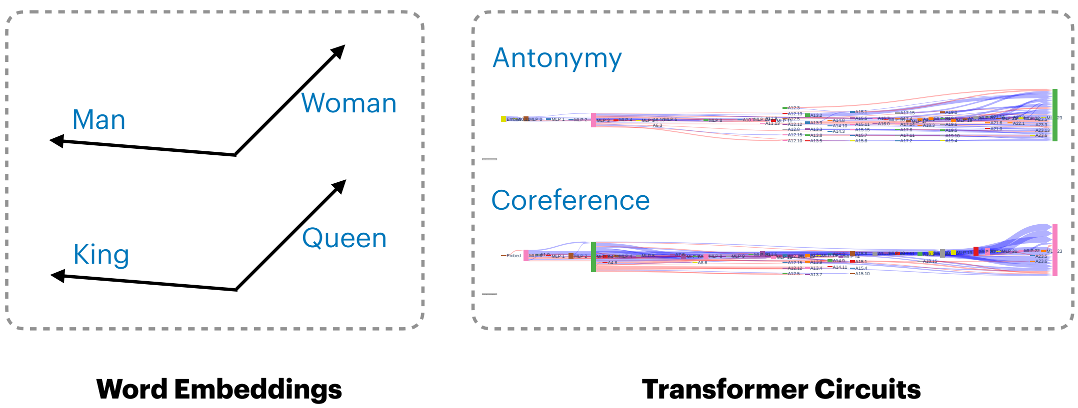

Figure 1: Comparison of word embeddings (left) and transformer circuits (right).

Word embeddings represent *meanings*. We are familiar with their geometric properties, for example $vec(man) - vec(woman) = vec(king) - vec(woman)$ (Figure 1).

Transformer circuits represent *functions*. They are sparse computational subgraphs within a transformer that approximate the full model [MCS24]. However, little is known about their relationships. For instance, if we combine $circuit(antonymy) + circuit(coreference)$, does the combined circuit capture coreference with an emphasis on contrasting meaning? In this project, we propose constructing a "safari" of transformer circuits and developing a more precise description of circuit arithmetic.

This proposal is also available at `https://yisong.me/topics/CircuitSafari.pdf` and will remain online until the position is filled (after which we will switch to stealth mode).

2 Approach

Our work addresses two major challenges in the transformer circuits community:

Challenge 1: Limited language (and circuit) resources. Circuit discoveries have so far been made in isolation by different teams. An incomplete list includes indirect object identification (IOI), numerical comparison, subject-verb agreement (SVA), MCQ, knowledge acquisition, colored objects, extractive QA, context-free grammars, and discourse relations [WVC+23, HLV23, FCj24, LRK+23, YZX+24, OYZ+25, HPB24, MEP24, BMR+25, MWP25, MK25]. We aim to unify these efforts and extend coverage to the "classic NLP pipeline" [TDP19], filling gaps such as syntactic features, semantic features, and semantic role labeling. To this end, we will collect the necessary language resources to support circuit discovery and release the resulting circuits.

Challenge 2: Limited understanding of circuit arithmetic. Prior work has explored circuit composition [MWP25], but with limitations: (1) it focuses on synthetic probabilistic context-free grammar, detached from real-world language; (2) it examines simple set union as composition. To address (1), we will conduct systematic experiments across our Safari. To address (2), we will investigate richer arithmetic operations, such as edge subtraction or weighted set unions, and explore making composition differentiable for end-to-end optimization.

3 Organization

Who will I be working with? In this project, you will be working with WING PhD student Yisong Miao and WING lead Prof. Min-Yen Kan.

When is the starting date? January 2026 or earlier (e.g. October 2025).

What are the prerequisites? Strong natural language processing (NLP) experience, e.g., you have led a team to excel in the CS4248 / CS6101 project components (or equivalent).

What are the expected commitments and outcome? For UROP/FYP/MComp students, we expect sufficient time commitment to achieve excellence in this project. The expected outcome is a research publication, so applicants with academic inclinations are preferred.

How to get in touch? Please email Yisong and Prof. Min. We also copy Min's message: "You *must* meet with me on Zoom or via email before deciding to take this project or any other project under my group. Please email me first to arrange a suitable timing. If you bid without consulting me, I will refuse your project bid. Read my notes on projects (`http://www.comp.nus.edu.sg/~kanmy/hyp.html`) and learn about the host organization, WING `http://wing.comp.nus.edu.sg/`, and read the requirements about entry to WING: `http://bit.ly/new-to-wing` (there's a narrated video at `http://bit.ly/new-to-wing-yt` as well)." This project is primarily supervised by Yisong, a doctoral student under Min. Min will be overseeing the project, but not necessarily working directly with you.

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